SoftMarginLoss#

https://pytorch.org/docs/stable/generated/torch.nn.SoftMarginLoss.html

Description:

Creates a criterion that optimizes a two-class classification logistic loss between input tensor xxx and target tensor yyy (containing 1 or -1). size_average (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field size_average is set to False, the losses are instead summed for each minibatch. Ignored when reduce is False. Default: True reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size_average. When reduce is False, returns a loss per batch element instead and ignores size_average. Default: True reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean' Input: $(*)$(*)$(*)$, w

Function Signature

```
class torch.nn.SoftMarginLoss(size_average=None,
reduce=None, reduction='mean')[source]#

Parameters

Shape:

forward(input, target)[source]#

Return type
```

Returns:

Tensor

Detailed Documentation

Documentation

Creates a criterion that optimizes a two-class classification logistic loss between input tensor xxx and target tensor yyy (containing 1 or -1).

size_average (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field size_average is set to False, the losses are instead summed for each minibatch. Ignored when reduce is False. Default: True

reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size_average. When reduce is False, returns a loss per batch element instead and ignores size_average. Default: True

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

Input: $(*)$(*)$(*)$, where $*$*$*$ means any number of dimensions.

Target: $(*)$(*)$(*)$, same shape as the input.

Output: scalar. If reduction is 'none', then $(*)$(*)$(*)$, same shape as input.

Runs the forward pass.